

Molecular Graph Captioning

ALTEGRAD Data Challenge

Team EAG

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- **Molecular Graph Captioning:** generating natural language descriptions from molecular graphs
- Our focus: a **retrieval-based approach**, with additional generative and hybrid approaches

- **Dataset**

- ~33k molecular graphs with natural language descriptions
- Splits: 31,008 train / 1,000 val / 1,000 test (captions hidden)
- Stored as PyTorch Geometric Data objects

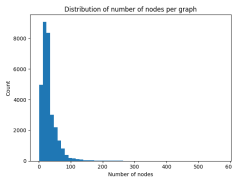
- **Graph representation**

- Nodes = atoms, edges = chemical bonds
- **Node features:** 9 categorical attributes (atom type, degree, charge, hybridization, aromaticity, ring membership, ...)
- **Edge features:** bond type, stereochemistry, conjugation

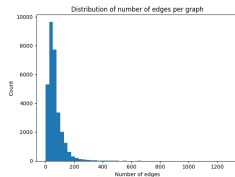
Data - Graph and Caption Statistics

● Graph structure

- Nodes = atoms, edges = chemical bonds
- Atoms: 1 to 574 (mean ~ 32)
- Bonds: 0 to 1,284 (mean ~ 67)



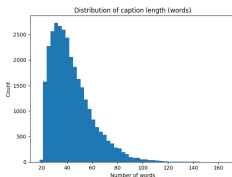
Nodes per molecule



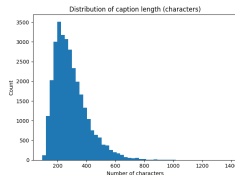
Edges per molecule

● Non-uniform distributions

- Many small molecules
- Long tail of large graphs
- Similar across all splits



Caption length (words)



Caption length
(characters)

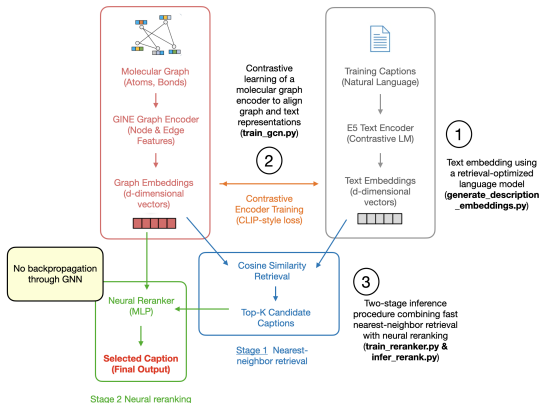
● Caption variability

- 18–166 words (mean ~ 43)
- 95–1,377 characters (mean ~ 294)

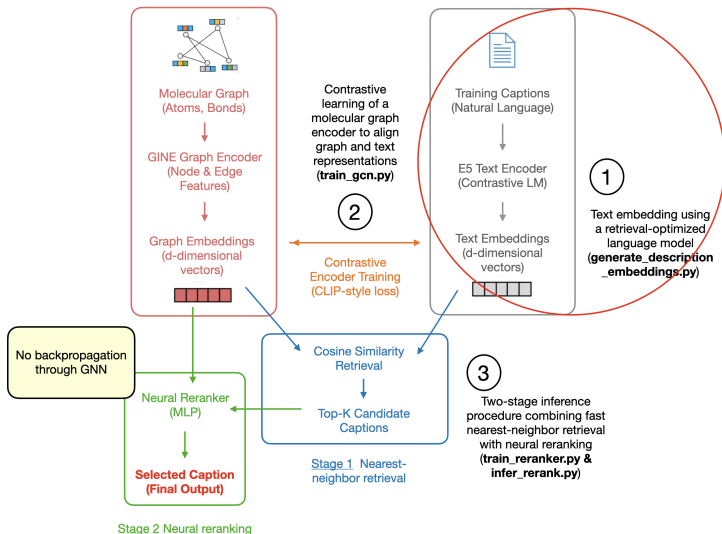
Retrieval-Based Method

Retrieval-Based Method (1) - High-Level Idea

- **Retrieval-based molecular graph captioning**
- Retrieve the most relevant caption in a shared graph-text embedding space
- Similarity measured with cosine similarity



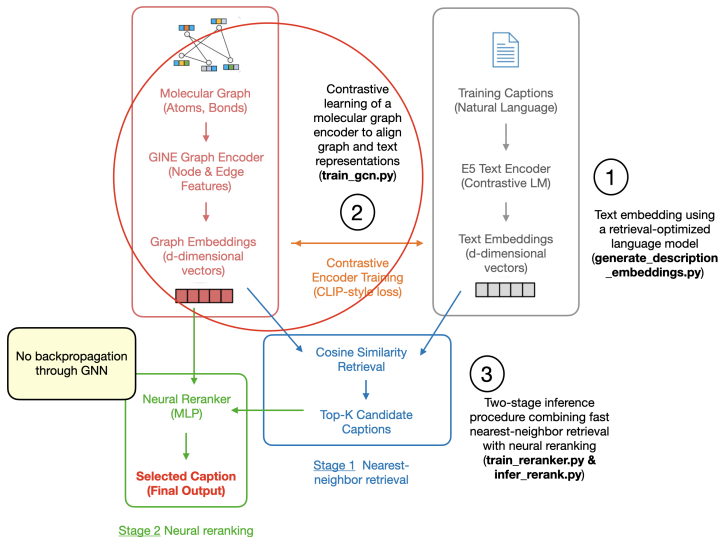
Retrieval-Based Method (1) - Text Representation



Retrieval-Based Method (1) - Text Representation

- Baseline: BERT [CLS] embeddings
- Limitation: not optimized for similarity-based retrieval
- We use `intfloat/e5-base-v2`
 - Contrastively trained for dense retrieval
 - Better cosine similarity structure
- Mean pooling + ℓ_2 normalization
- Text embeddings are **precomputed and frozen**

Retrieval-Based Method (1) - Molecular Graph Encoder



Retrieval-Based Method (1) - Molecular Graph Encoder

- Baseline: simple GCN without atom / bond features
- Limited chemical expressiveness
- We use a **GINE** architecture (Graph Isomorphism Network with Edge features)
- Explicitly incorporates:
 - atom features
 - bond features (type, stereochemistry, conjugation)
- Stacked GINE layers with residual connections and layer normalization
- Attention-based pooling for graph-level embeddings
- Projection head + ℓ_2 normalization

Retrieval-Based Method (1) - Contrastive Graph-Text Alignment

- Baseline uses MSE loss $\rightarrow$ weak separation of negatives
- We adopt a **CLIP-style contrastive loss**

$$\mathcal{L}_i = -\log \frac{\exp(\text{scale } \langle g_i, t_i \rangle)}{\sum_{t \in \{t_1, \dots, t_B\} \cup \mathcal{Q}} \exp(\text{scale } \langle g_i, t \rangle)}$$

- Memory queue $\mathcal{Q}$ increases number of negatives
- Learnable temperature controls similarity sharpness

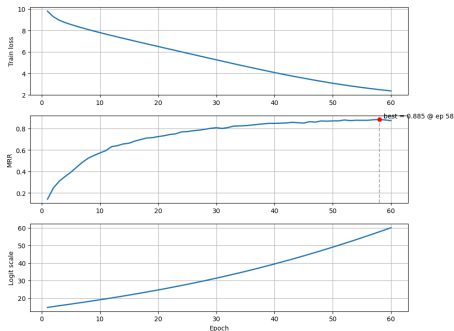
Retrieval-Based Method (1) - Training and Model Selection

- Graph encoder trained with contrastive loss
- Validation set used only for model selection
- Evaluation metric: **Mean Reciprocal Rank (MRR)**

$$\text{MRR} = \frac{1}{N} \sum_{i=1}^N \frac{1}{r_i}$$

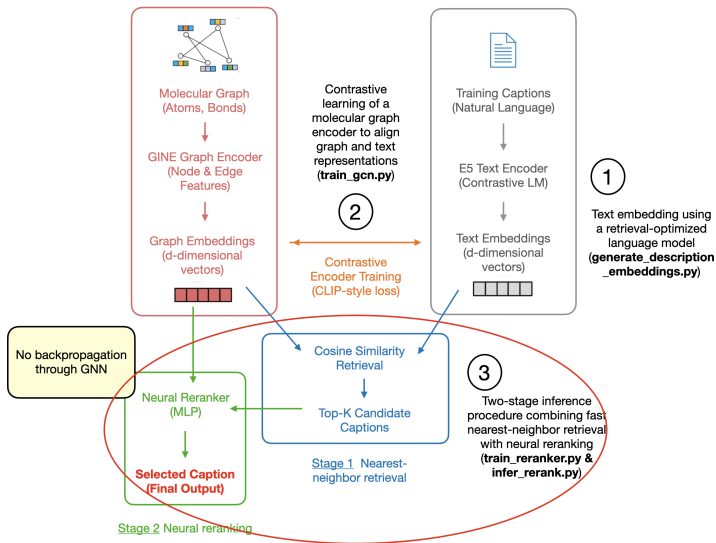
- Training details:
 - 5 GINE layers, hidden dim 512
 - AdamW, lr 2×10^{-4} , batch size 256, Memory queue of 65k text embeddings

Training dynamics of the contrastive GNN encoder



Training dynamics of the contrastive graph encoder

Retrieval-Based Method (1) - Two-Stage Inference



Retrieval-Based Method (1) - Two-Stage Inference

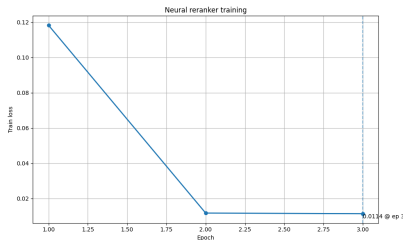
- **Stage 1: Retrieval**

- Embed query graph
- Retrieve top- K captions by cosine similarity

- **Stage 2: Neural reranking**

- Reranker trained on frozen embeddings
- Listwise cross-entropy loss
- Improves fine-grained ranking among similar captions

- Final caption = highest reranker score



Training loss of the neural reranker

- **Evaluation (Kaggle)**

- Composite metric:
 - BLEU-4 F1 (lexical overlap)
 - BERTScore (semantic similarity, RoBERTa)
- Final score aggregates lexical + semantic similarity

- **Leaderboard performance**

- Baseline (provided): ≈ 0.487 (public)
- Our retrieval pipeline (contrastive + reranking):
 - **0.661** (public leaderboard)
 - **0.682** (private leaderboard)

- **Why does it work?**

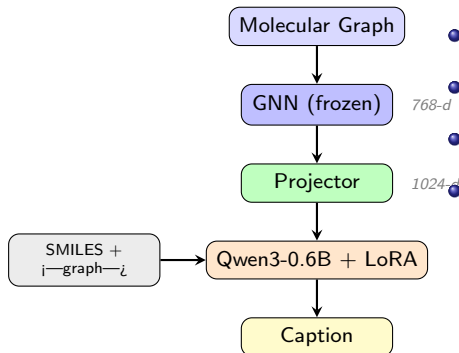
- Retrieval-optimized text embeddings (E5)
- Chemically expressive graph encoder (GINE + edge features)
- Contrastive alignment with many hard negatives (memory queue)
- Two-stage inference with neural reranking

End-to-End Generation

End-to-End Generation (2) - Overview

- **Goal:** Generate novel captions directly from molecular graphs
 - Not restricted to training corpus captions
- **Architecture:** GNN $\rightarrow$ Projector $\rightarrow$ LLM
 - Reuse frozen GNN encoder from retrieval
 - Fine-tune LLM with LoRA adapters
- **Key challenge:** Limited compute (Google Colab free tier)

End-to-End Generation (2) - Architecture



- **Graph Encoder:** Frozen GINE (768-d)

- **Projector:** 3-layer MLP (768→1024)

- **LLM:** Qwen3-0.6B with LoRA ($r=64$)

- **Single-Token Injection**

- Graph → **one** `<|graph|>` token
- Unlike vision: no spatial patches
- Retrieval validates: single embedding captures graph semantics

End-to-End Generation (2) - Input Format

SMILES Representation

- Text-based molecular encoding
- ASCII: atoms, bonds, connectivity

Example:

Nc1ncnc2c1ncn2C1OC(COP(=O)([O-]))...

Token Injection

- `<|graph|>` embedding replaced with projected graph
- LLM sees both text + graph

Chat Template

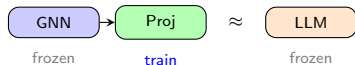
```
<|im_start|>system
You are an expert chemist. Describe the
molecule's properties concisely.<|im_end|>
<|im_start|>user
Molecule Structure: <|graph|>
SMILES: NC(=NCCCC(N)C(=O)O)NP(=O)(O)O<|im_end|>
<|im_start|>assistant
<think>
</think>
The molecule is an arginine derivative...
```

Future: Chain-of-thought via `<think>` tags for complex molecules

End-to-End Generation (2) - Two-Stage Training

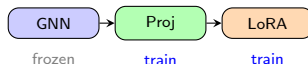
Stage 1: Alignment

- **Loss:** MSE (graph $\rightarrow$ text embedding)
- **Train:** Projector only
- **Freeze:** GNN + LLM
- **LR:** 10^{-3}



Stage 2: Supervised Fine-Tuning

- **Loss:** Cross-entropy (captions)
- **Train:** Projector + LoRA
- **Freeze:** GNN + LLM base
- **LR:** 10^{-4} / 10^{-5} (LoRA)



End-to-End Generation (2) - Generated Examples

Example: Amino tetrasaccharide

Ground truth: "The molecule is an amino tetrasaccharide comprised of an L-iduronic acid residue, sulfated on O-2, a glucuronic acid residue, and two N-acetylated galactosamine residues, one of which is at the reducing end and both of which are sulfated on O-4. It is an intermediate in the chondroitin sulfate degradation pathway. It has a role as a mouse metabolite."

Generated: "The molecule is an oligosaccharide acid that is the 4-deoxy derivative of the sulfate acid derivative of N-acetylglucosamine."

Example: Neolignan glucoside

Ground truth: "The molecule is a neolignan isolated from the whole plant of *Bidens parviflora* that has been found to inhibit histamine release from the peritoneal exudate mast cells induced by antigen-antibody reaction. It has a role as a metabolite and a histamine antagonist. It is a beta-D-glucoside, an aromatic ether, a neolignan, a polyphenol, a diol and a secondary alcohol."

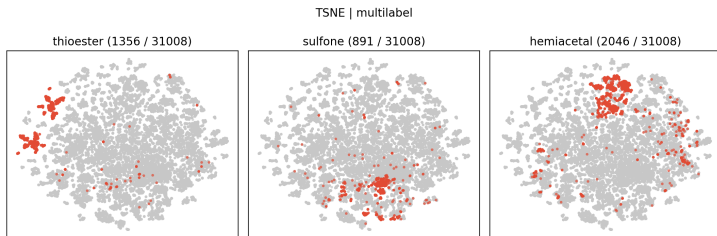
Generated: "The molecule is a member of the class of dihydroxybenzoic acids and a 2-hydroxybenzoic acid. It is a member of dihydroxybenzoic acids and a 2-hydroxybenzoic acid."

Observations:

- ✓ Learns correct format and chemical vocabulary
- ✗ Some hallucination, repetition in outputs (undertrained)

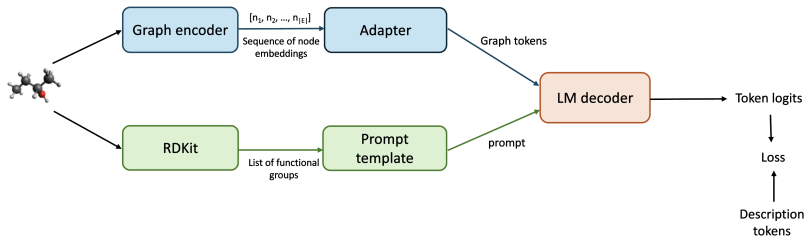
Retrieval-Augmented Generation

Retrieval-Augmented Generation - Functional Group Prompting (3)



- graph embeddings show functional group clusters
- enhance generation with (very compact!) prompting
- coarse-grained interpretable/stable conditioning

Retrieval-Augmented Generation - Functional Group Prompting (3)

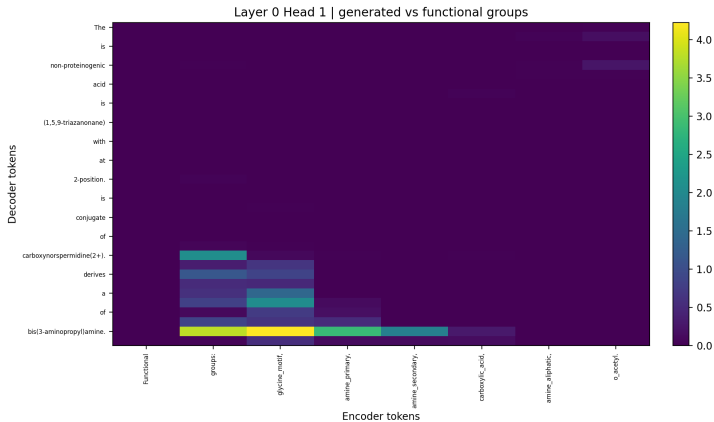


- GINE encoder is frozen
- Only 6 rarest functional groups are included in the prompt

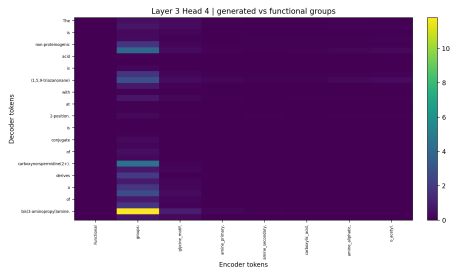
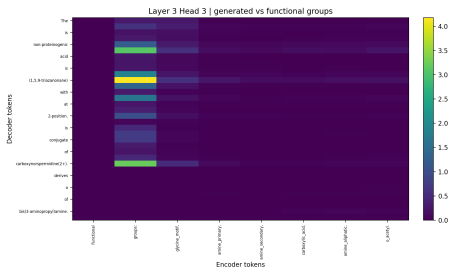
Retrieval-Augmented Generation - Functional Group Prompting (3)



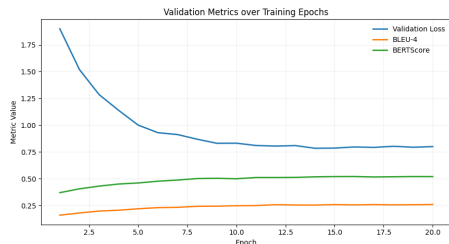
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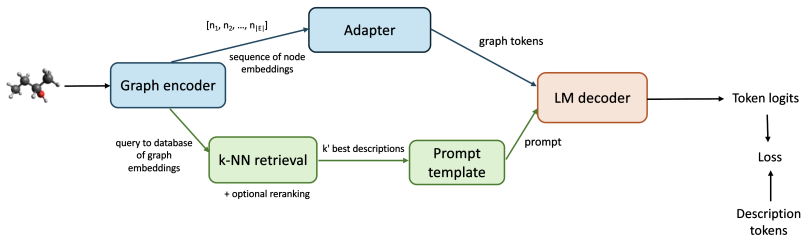
Example: Amino tetrasaccharide

Ground truth: "The molecule is an amino tetrasaccharide comprised of an L-iduronic acid residue, sulfated on O-2, a glucuronic acid residue, and two N-acetylated galactosamine residues, one of which is at the reducing end and both of which are sulfated on O-4. It is an intermediate in the chondroitin sulfate degradation pathway. It has a role as a mouse metabolite."

Generated: "" The molecule is an amino trisaccharide consisting of N-acetyl-beta-D-galactosaminyl, 2-acetamido-2-deoxy-alpha-L-threo-hex-4-enopyranuronic acid and an oligosaccharboxylic acid residue joined in sequence by a (1-4) glycosidic bond."

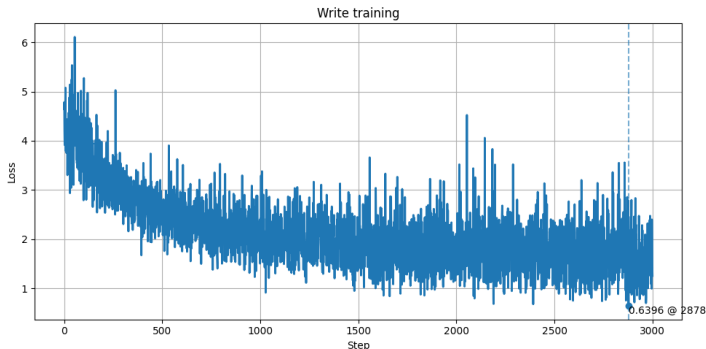
- Stable training
- Generated captions are descent

Retrieval-Augmented Generation - Top-K Description Prompting (4)



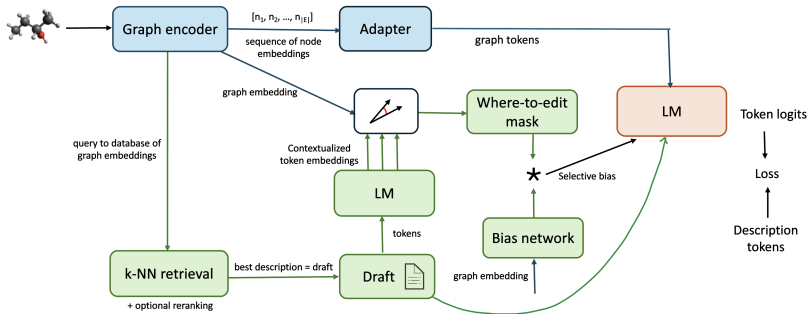
- Prompting makes the training very compute-intensive (it scales quadratically!)
- More fine-grained prompt, but may contain misleading facts...

Retrieval-Augmented Generation - Top-K Description Prompting (4)



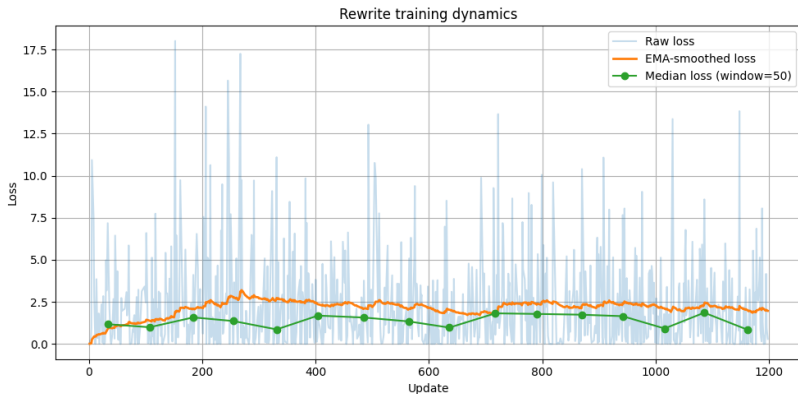
- The loss decreases steadily
- Conditioning on prompt and graph prefix seems to work
- Very small training due to computing limitations

Retrieval-Augmented Generation - Retrieval-Guided Description Rewriting (5)



- Graph encoder is frozen, LMs are frozen. We train adapter and bias network.
- Cross-entropy loss is computed on the where-to-edit tokens

Retrieval-Augmented Generation - Retrieval-Guided Description Rewriting (5)



- Loss is very noisy
- Model seems to be adapting to the new loss signal at the beginning of the training then stabilizes

Generative-Based Results

Method	Public score
Baseline (0)	0.487
Retrieval (no reranking)	0.604
Retrieval (with reranking) (1)	0.661
End-to-End Generation From Scratch (2)	0.507
Functional-Group Prompting (3)	0.491
Top-K Description Prompting (4)	0.436
Retrieval-Guided Description Rewriting (5)	0.594

Table: Public Kaggle leaderboard scores for the different approaches.

Limitations (generation)

- Google Colab free tier
- Session disconnections
- Could not run full training ($\sim 9h$)

Not fundamental

- Architecture is sound
- More compute would help

Conclusion

- Retrieval-based methods dominate under limited compute
- Generative and RAG approaches are constrained by GPU availability
- More GPU budget would enable longer training and stronger generation

Thank you for your attention.